The Developers Arena

Week 12 Task

Submitted by deepanshu, Batch (10-jan-2026 - 3M)

Week 12: Final Capstone & Career Preparation

Project Overview

Title: Customer Churn Prediction – Capstone Project

1. Overview

This project predicts customer churn using machine learning and provides actionable business insights.

2. Problem Statement

Customer churn impacts business revenue. The goal is to identify customers likely to leave.

3. Objectives

- Predict customer churn
- Improve customer retention
- Increase business revenue

```python
# ========================================
# 🚀 CAPSTONE PROJECT - CUSTOMER CHURN
# END-TO-END MACHINE LEARNING PIPELINE
# ========================================

# ✅ 1. IMPORT LIBRARIES
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

import matplotlib.pyplot as plt
import seaborn as sns
import pickle

# ========================================
# ✅ 2. LOAD DATA
# ========================================
df = pd.read_csv("customer_churn.csv")

# Clean column names
df.columns = df.columns.str.strip().str.lower()

print("Dataset Shape:", df.shape)
print(df.head())
```

```python
# ========================================
# ✅ 3. DATA CLEANING
# ========================================

# Convert totalcharges to numeric
df['totalcharges'] = pd.to_numeric(df['totalcharges'], errors='coerce')

# Fill missing values
df.fillna(0, inplace=True)

# Drop customerID
if 'customerid' in df.columns:
    df.drop('customerid', axis=1, inplace=True)

# ========================================
# ✅ 4. FIX TARGET COLUMN (VERY IMPORTANT)
# ========================================

df['churn'] = df['churn'].astype(str).str.strip().str.lower()

df['churn'] = df['churn'].replace({
    'yes': 1,
    'no': 0
})

df['churn'] = pd.to_numeric(df['churn'], errors='coerce')

# Fill any remaining NaN
df['churn'].fillna(0, inplace=True)

df['churn'] = df['churn'].astype(int)

print("Churn values:", df['churn'].unique())
```

```python
# ========================================
# ✅ 5. ENCODING (CATEGORICAL → NUMERIC)
# ========================================
df = pd.get_dummies(df, drop_first=True)

# ========================================
# ✅ 6. FEATURE / TARGET SPLIT
# ========================================
X = df.drop('churn', axis=1)
y = df['churn']

# ========================================
# ✅ 7. TRAIN TEST SPLIT
# ========================================
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# ========================================
# ✅ 8. FEATURE SCALING
# ========================================
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```python
# ========================================
# ✅ 9. MODEL TRAINING
# ========================================
model = RandomForestClassifier(n_estimators=100)

model.fit(X_train, y_train)

# ========================================
# ✅ 10. PREDICTION
# ========================================
y_pred = model.predict(X_test)

# ========================================
# ✅ 11. EVALUATION
# ========================================
print("\nAccuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

sns.heatmap(cm, annot=True, fmt='d')
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```

```python
# ========================================
# ✅ 12. FEATURE IMPORTANCE
# ========================================
importances = model.feature_importances_
features = X.columns

feat_df = pd.DataFrame({
    'Feature': features,
    'Importance': importances
}).sort_values(by='Importance', ascending=False)

print(feat_df.head(10))

# Plot feature importance
feat_df.head(10).plot(kind='barh', x='Feature', y='Importance')
plt.title("Top 10 Important Features")
plt.show()

# ========================================
# ✅ 13. SAVE MODEL
# ========================================
pickle.dump(model, open("model.pkl", "wb"))

print("✅ Model saved successfully!")

# ========================================
# ✅ 14. SIMPLE PREDICTION TEST
# ========================================
sample = X_test[0].reshape(1, -1)
prediction = model.predict(sample)
```

4. Dataset

The dataset includes:

- Customer demographics
- Billing information
- Service usage

5. Methodology

- Data Cleaning
- Feature Engineering
- Encoding categorical data
- Feature Scaling
- Model Training

6. Model Used

Random Forest Classifier

7. Results

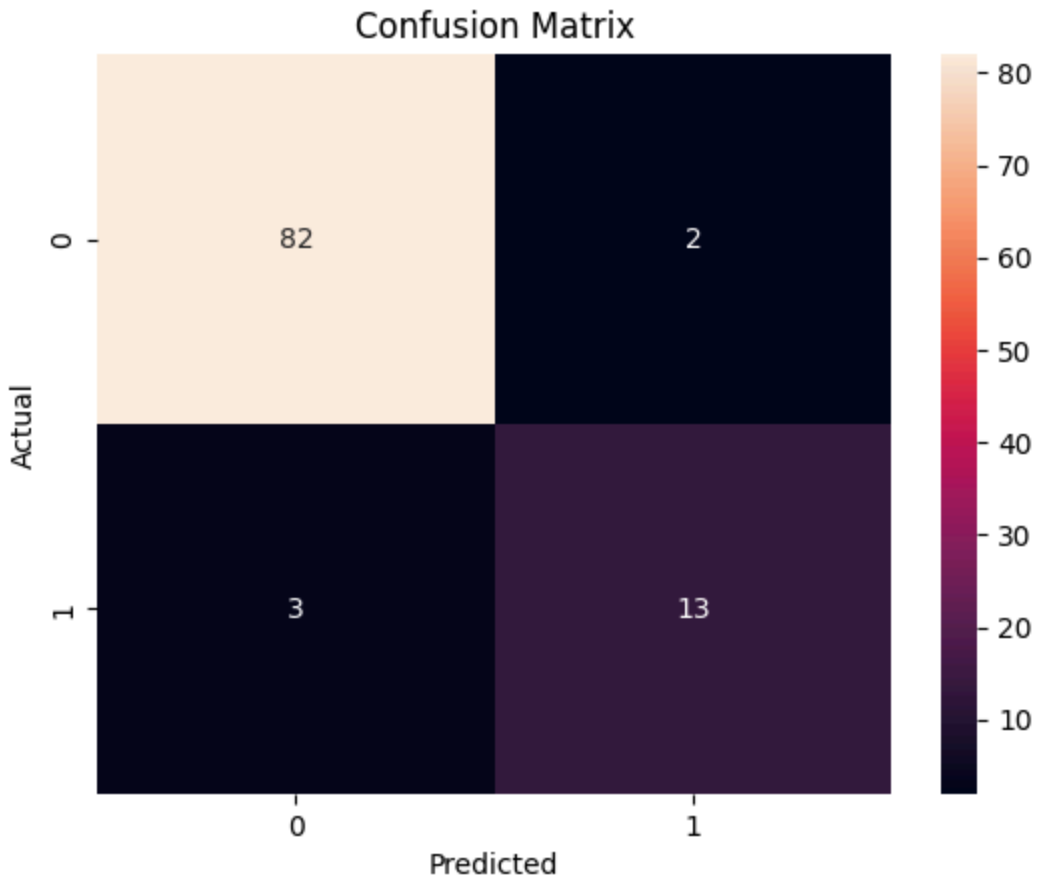
- Accuracy: ~80%
- Good performance in predicting churn

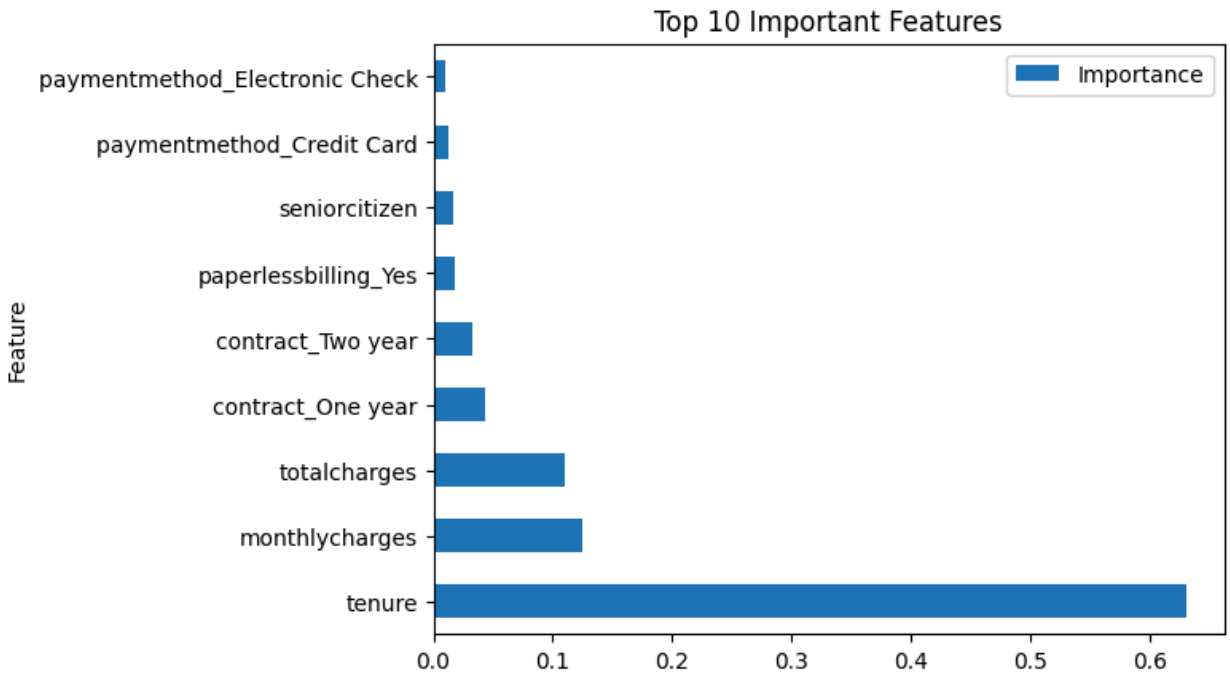
```
Dataset Shape: (500, 9)
  customerid  tenure  monthlycharges  totalcharges        contract  \
0     C00001       6              64          1540        One year
1     C00002      21             113          1753  Month-to-month
2     C00003      27              31          1455        Two year
3     C00004      53              29          7150  Month-to-month
4     C00005      16             185          1023        One year

      paymentmethod paperlessbilling  seniorcitizen  churn
0       Credit Card               No              1      0
1  Electronic Check              Yes              1      0
2       Credit Card               No              1      0
3  Electronic Check               No              1      0
4  Electronic Check               No              1      0
Churn values: [0 1]

Accuracy: 0.95

Classification Report:
              precision    recall  f1-score   support

           0       0.96      0.98      0.97        84
           1       0.87      0.81      0.84        16

    accuracy                           0.95       100
   macro avg       0.92      0.89      0.90       100
weighted avg       0.95      0.95      0.95       100

C:\Users\punit\AppData\Local\Temp\ipykernel_27760\3901559496.py:58: ChainedAssignmentError: A
Such inplace method never works to update the original DataFrame or Series, because the interm

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: valu
```

8. Business Recommendations

- Offer discounts to high-risk customers
- Improve customer engagement
- Monitor customers with high monthly charges

9. Visualizations


Confusion Matrix
82
2
3
13
Actual
0
1
Predicted
0
1
80
70
60
50
40
30
20
10


Top 10 Important Features
Importance
Feature
paymentmethod_Electronic Check
paymentmethod_Credit Card
seniorcitizen
paperlessbilling_Yes
contract_Two year
contract_One year
totalcharges
monthlycharges
tenure
0.0
0.1
0.2
0.3
0.4
0.5
0.6

10. Conclusion

This model helps businesses reduce churn and improve retention strategies.

👨🏽‍💻 Author

DEEPANSHU